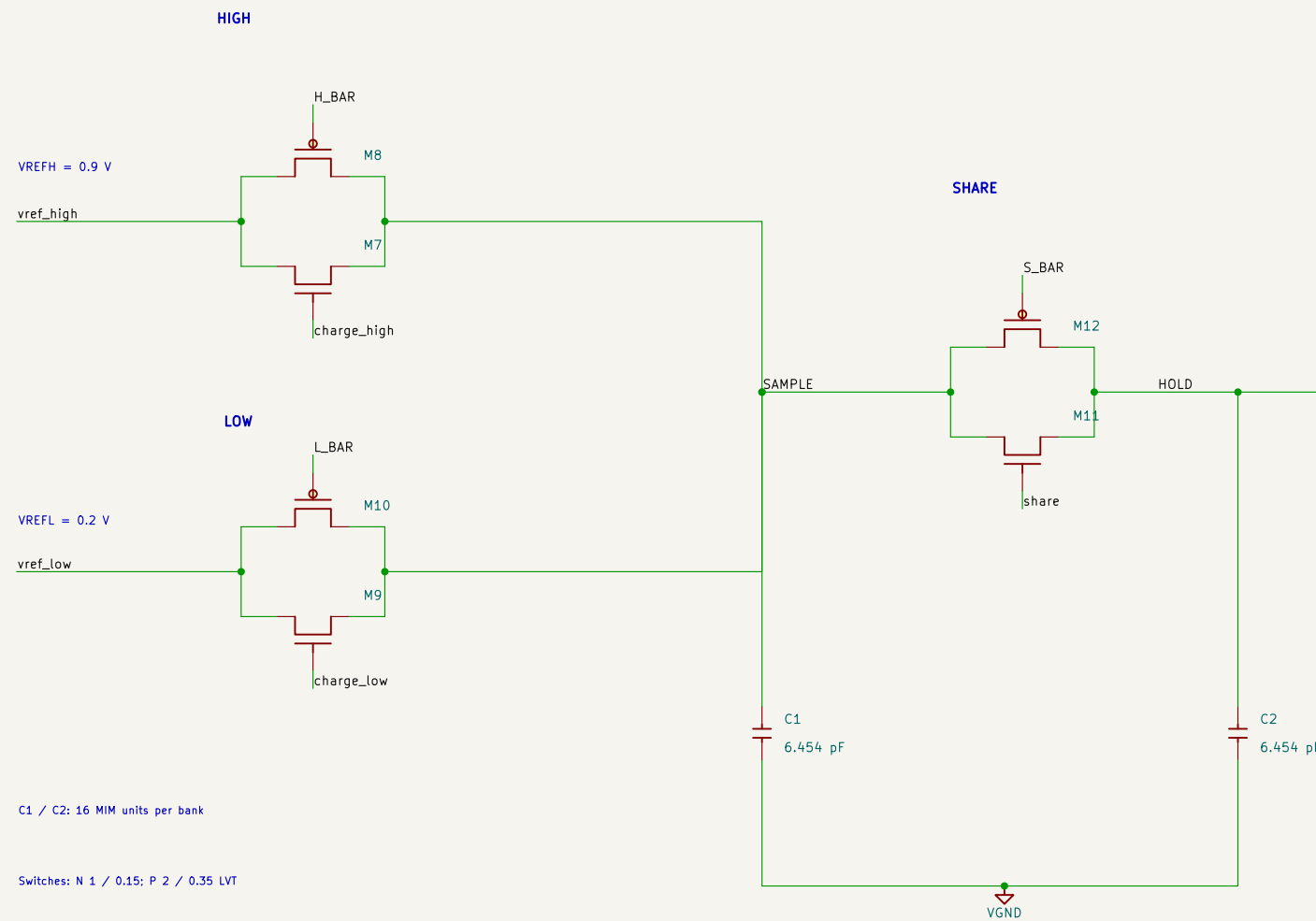


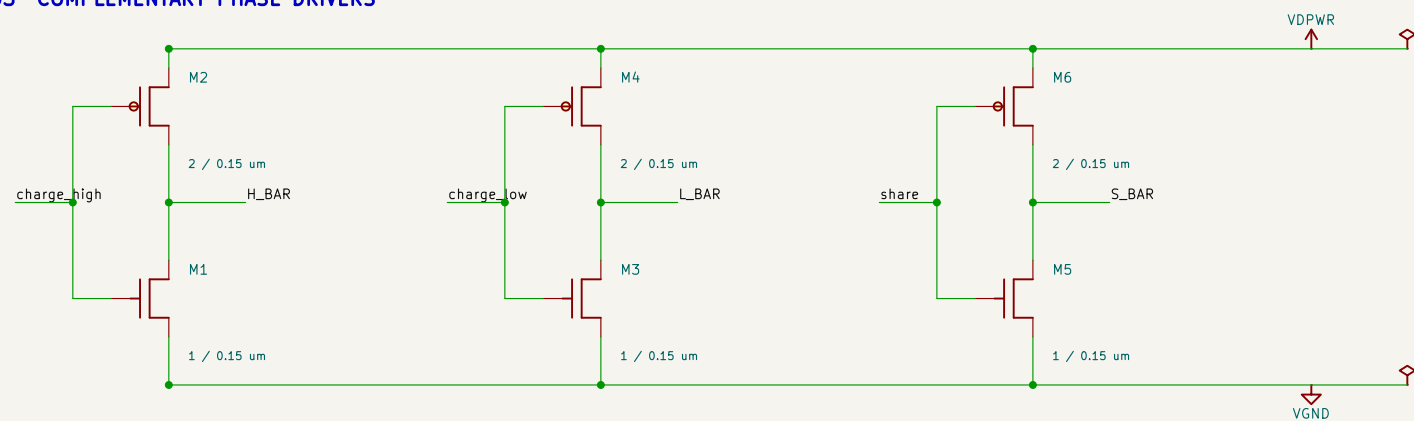
SUAREZ SERIAL CHARGE-SHARING DAC

SKY130 | 1.8 V | 8 bits, LSB first | 180 x 115 um | W / L dimensions in um

01 REFERENCE SELECTION AND CHARGE SHARING



03 COMPLEMENTARY PHASE DRIVERS



OPERATING SEQUENCE

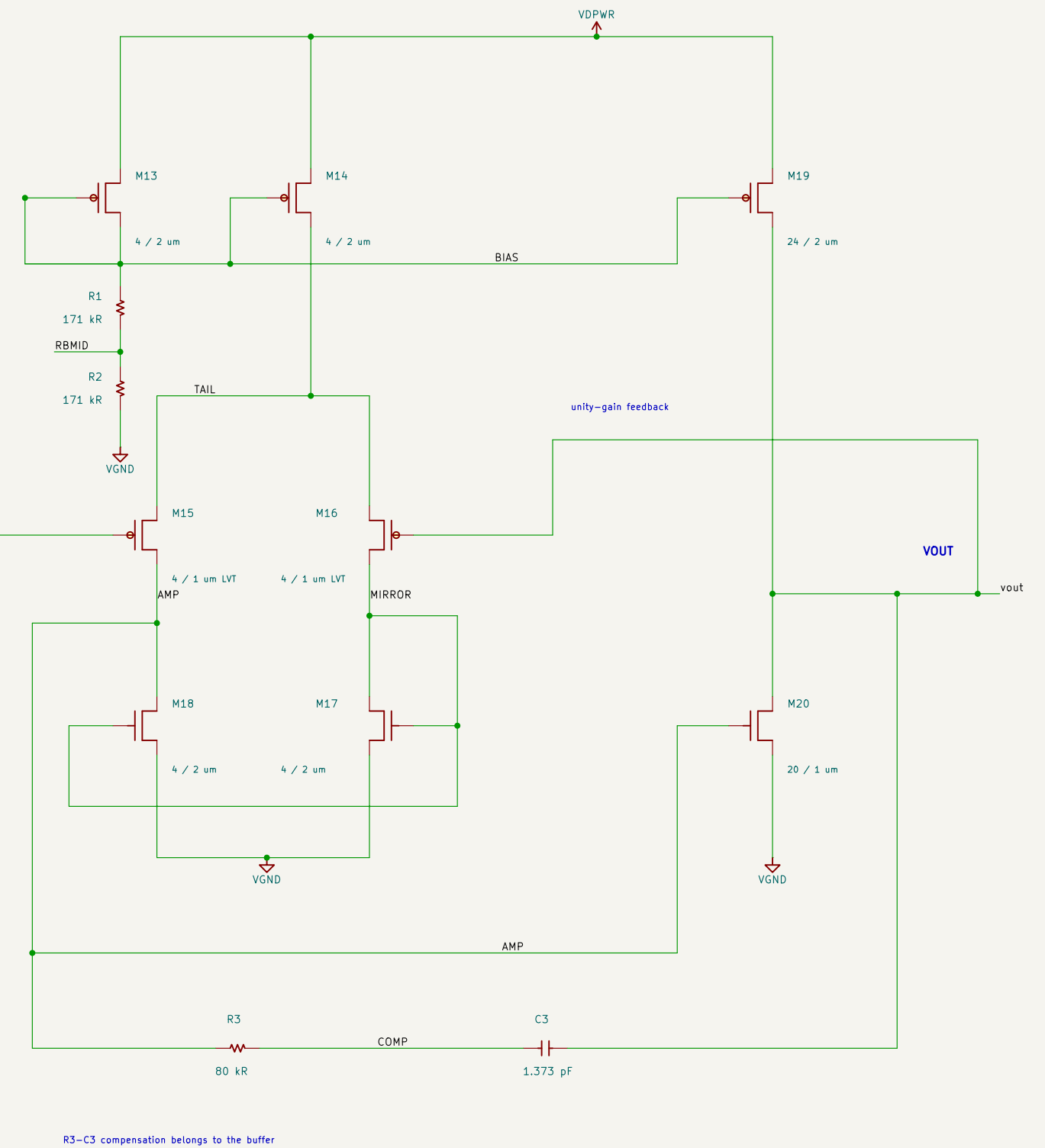
Initialize: HIGH=0, LOW=1, SHARE=1 for 8 us. Then all phases off for 0.2 us.

8 bits, LSB first: charge 2 us; dead 0.2 us; share 2 us; dead 0.2 us. Read 3 us after bit 7.

HIGH / LOW never overlap. LOW / SHARE overlap only during initialization. External controller supplies dead time.

Bodies: all NMOS $\rightarrow$ VGND; all PMOS $\rightarrow$ VDPWR. Body ties are explicit in SPICE / silicon.

02 UNITY-GAIN BUFFER / BIAS, FEEDBACK AND COMPENSATION



SKY130 / Siliccluster v3

Sheet: /
File: suarez_dac.kicad_sch

Title: Suarez two-capacitor DAC

Size: A3	Date:
KiCad E.D.A. 10.0.6	

Rev: 1.0
Id: 1/1